Case Study Report

Walking Towards Sustainability: Optimizing Public Transit though the “Walkable Cities” by Time-Oriented Model

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Abstract

Urban planning has changed in both real and intangible ways to address the pressing problems of sustainability, climate resilience, and efficient urban mobility. In order to make cities more livable, accessible, and connected, sustainable transportation ideas are heavily emphasized in contemporary urban planning techniques. One of the most crucial urban regulations that cities must adhere to in recent years is the idea of sustainable mobility. One of the most popular approaches for designing sustainable urban transportation is the "10-minute cities" model. Combating the detrimental effects of climate change on cities is another major function of this paradigm. This study examines the current public transportation stops' walking distances and their distances to one another from a methodological standpoint. The efficiency of the stops, their significance in multi-modal public transportation, which public transportation lines (particularly bus lines) they serve as they currently exist, and the number of passengers they serve are all determined by calculating the 5-8-10 minute walking circular areas to the stops. In addition to making public transportation more appealing by doing a walkability analysis of the stations, more effective stop locations can also be created by optimizing the stops that are inefficient or lack sufficient demand. With an average walking speed of 5 km/h, it is estimated that a person can cover 416 meters in 5 minutes, 666 meters in 8 minutes, and 832 meters in 10 minutes. Therefore, when an analysis is conducted based on these walking figures, the effects of the pauses on residential, industrial, and commercial districts may also be tracked. In addition to encouraging a more efficient and inclusive urban walkability system, this approach supports the development of flexible and sustainable urban landscapes. Therefore, the study's outputs can be very useful to urban planners who want to incorporate innovative and sustainable mobility solutions in their communities.

Keywords

Walkability, Geographic Information System, Time-Oriented Model, Public Transit, Sustainability

1. Introduction

To preserve and improve the standard of living in cities, sustainable urban mobility is essential. There are now more options and demands for public transit and transportation networks as the population of cities grows. Urban transportation planning has therefore emerged as a crucial issue for cities (Vuchic, 2017). According to the European Commission, sustainable mobility plans concentrate on creating urban transportation networks by fulfilling important objectives, such as providing transportation options for all

citizens, ensuring safety, combatting pollution, being cost-effective, and enhancing the standard of living in urban areas (Chou, 2017). To satisfy the needs of passengers and citizens, administrators create a variety of transportation projects and solutions. These initiatives and solutions are carried out in accordance with specific aims and objectives. When waiting or switching between different forms of transportation, passengers use stops. (Rietveld, 2004; García-Palomares, Gutiérrez and Cardozo, 2013; Johar, Jain and Garg, 2017; Jie, Zhiwei and Meng, 2019) Crucial metrics include the stops' location, passenger density, area served, and the number of transportation lines and modes they can transfer from the stop serve.

For cities to achieve sustainable mobility goals, public transportation is essential, and the network's quality must be acceptable (Mohammed and Senadheera, 2022). Social contact (Kaufmann, Audikana, and Drevon, 2021), energy efficiency (Moriarty and Honnery, 2012), and a sustainable environment (Tezangi, 2014) are all impacted by public transportation, both directly and indirectly. By implementing comprehensive public transit, cities may provide all passengers safer, more effective, and low-emission travel options (Kandt et al., 2015; Friman, Lättman, and Olsson, 2020).

Additionally, by offering accessible routes to transit terminals, public transportation systems can promote active transportation and encourage walking and bicycling as essential parts of daily commutes (Wigan, 1995) + Corazza & Favaretto, 2019; Lachapelle et al., 2011). Research shows that cities that make investments in multimodal transportation systems—where walking and bicycling infrastructure is seamlessly integrated with public transportation—have lower levels of vehicle dependency and higher rates of public transportation use (Rifai & Arifin, 2020). In addition to increasing the transport network's efficiency, this integration promotes an active mobility culture, which results in healthier lifestyles and less traffic in cities (Oleśków-Szłapka et al., 2020). Public transportation can lessen the cost burden of car ownership by decreasing the need for private vehicles, hence increasing mobility for low-income groups (Tiwari et al., 2021).

In this study, Geographic Information Systems (GIS) were used to analyze the walkability of the planned tram stops in Sakarya province, Turkey. Sakarya, which has a population of 1.098.115 and a surface area of 4817 km2, is situated in northwest Turkey (T.C. Sakarya Valiligi İl Brifingi, 2024) (Figure 1). Sakarya Metropolitan Municipality revealed the planned tram project, which is divided into three sections (Figure 2). After assessing the city's main transportation hubs at the beginning of the project, the authors decided that Phase 1 and the eastern portion of Phase 3 should be developed jointly, as seen in Figure 2. Based on this integrated route, a walkability analysis was carried out. The tramway project, which nearly matched the path taken in the authors' study, was approved after discussions between the municipality and the Ministry of Transport. The study's recommended route and the official plan's overlap have been approved as the designated infrastructure project (Figure 3).

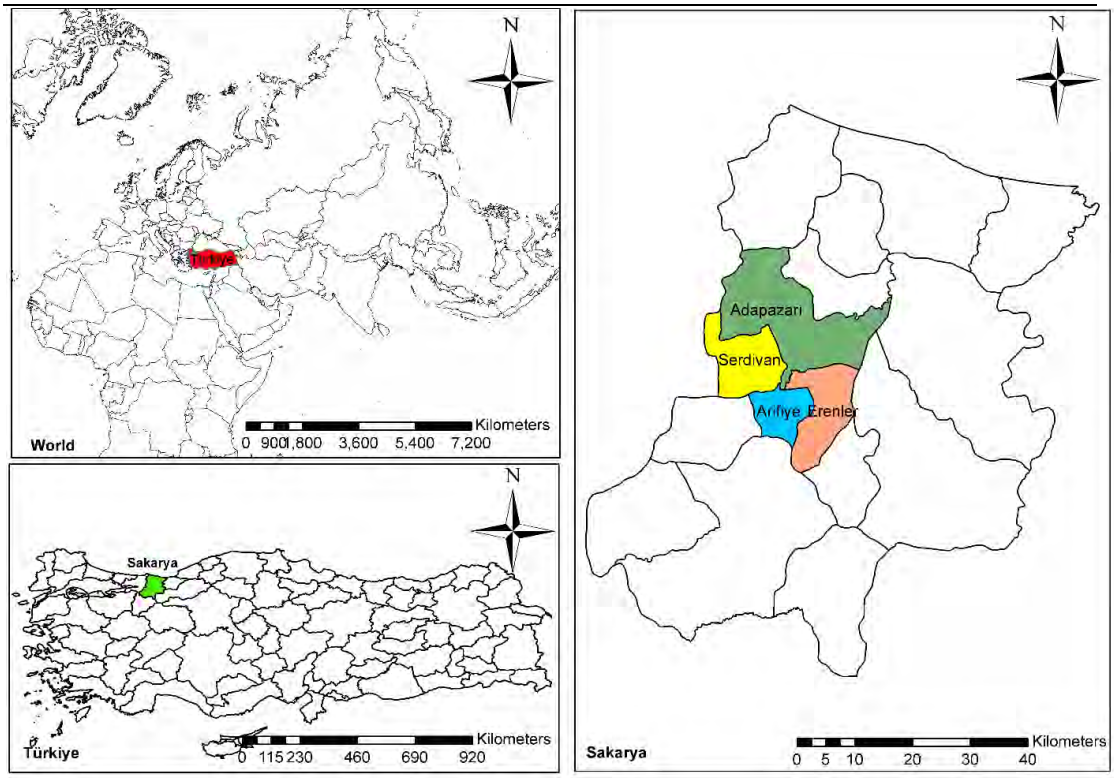

Figure 1. Location of Sakarya from various scales

Figure 2. The Planned Tram Project (*Raylı sistemde güzergahlar belirlendi*, 2023)

Figure 3. Analyzed tram lines and stops

2. Methodology and Analysis

The proposed tram line project will provide a new form of transportation to Sakarya, a city whose residents have long walked, cycled, and used train systems, all of which have influenced their travel habits. The new tram system is anticipated to significantly improve walkability and support current transit patterns, and residents are likely to swiftly adjust to it. In order to assess the accessibility of planned tram stops and their closeness to existing bus stops, walkable zones have been meticulously mapped. After the tram is finished, this analysis offers suggestions for improving bus routes, resulting in a more accessible and integrated transportation system that fits well with the city's established travel habits.

To build the structure of this paper and to assist the structure of sentences both editing and proofreading, ChatGPT-4o is used. In general, for analysis, ESRI's ArcGISTM is used in this study to identify and assess walkable areas. Walkable routes (polyline layer) are incorporated into the algorithm to identify these walkable locations. OpenStreetMap, an open access mapping tool, is the source of this data. Then, using ESRI's ArcGISTM Network Analysis Tool, this polyline data is examined, and walkable regions are established by enclosing them with a predetermined range of durations and distances. The walking pace employed in this study is 5 km/h, which is typically regarded in the literature to be between 4.90 and 5.80 km/h (Arango Diaz, 2008). Then, walkable zones are determined by calculating walking distances of five, eight, and ten minutes. A 5-minute walk is 416 meters, an 8-minute walk is 666 meters, and a 10-minute walk is 832 m.

The following map (Figure 4) displays the projected tram line and tram stations. Red stops indicate potential

tram stops in the near future, whereas blue stops indicate planned tram stops. The tram route is 12 kilometers long, with stops spaced 0.35 to 1.8 kilometers apart (Table 1). Additionally, three virtual tram stops were built in consideration of the attraction centers in the areas where a large number of students congregate, as these may be a potential future extension of the tram stops. For those who can use these three virtual stations for transportation, they are also appealing locations.

Table 1. Length of The Planned Tram Line

From	To	Length of Tram Line (km)
SAÜ CAMPUS	KYK DORMS	1.4
KYK DORMS	FARAZA	1.8
FARAZA	MEDAR	1.1
MEDAR HOSPITAL	SERDIVAN MUNICIPALITY	1.3
SERDIVAN MUNICIPALITY	ASEM	0.8
ASEM	AGORA	0.8
AGORA	MILLET BAHCESI	0.7
MILLET BAHCESI	OFIS SANAT	0.6
OFIS SANAT	DEMOKRASI SQUARE	0.6
DEMOKRASI SQUARE	YENICAMI	0.35
YENICAMI	REHABILITATION	0.55
REHABILITATION	SCHOOLS	0.9
SCHOOLS	STADIUM	1.1
TOTAL:		**12**

Figure 4. Tram lines, tram stops and virtual tram stops

Figure 5 depicts the walking spaces, often known as time-oriented regions. The regions that are green, orange, and red represent the 0–5, 5-8, and 8–10-minute walking times, respectively. When using this kind of transportation, passengers may be more inclined to walk to the tram stops (Aksoy, 2022). The walkability map in relation to the tramway line was displayed using buffer analysis. Walkability to the stops was demonstrated using network analysis, and tram line attractiveness was demonstrated using buffer analysis. In network analysis, walkability was examined using pedestrians or appropriate paths (Figure 6). Figure 7 clearly illustrates the distinction between buffer area analysis and network (service) area analysis. To put it briefly, network analysis requires a layer to be used for analysis, whereas buffer analysis can be done without a layer (Aslan and Kocaman, 2018). In network analysis, a specific network layer must be created, and the attributes in that network must be appropriately provided for an accurate analysis, whereas in buffer analysis, determining the distance from a line or point is sufficient.

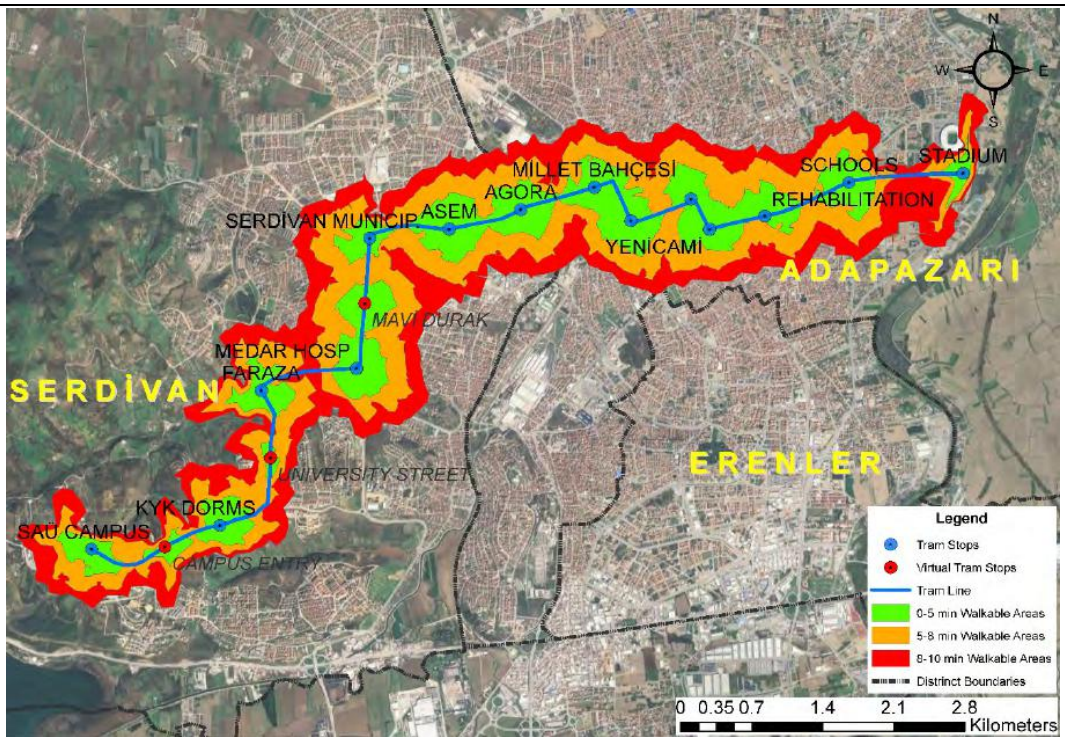

Figure 5. Time-Oriented Areas in 5-8-10 min to walk

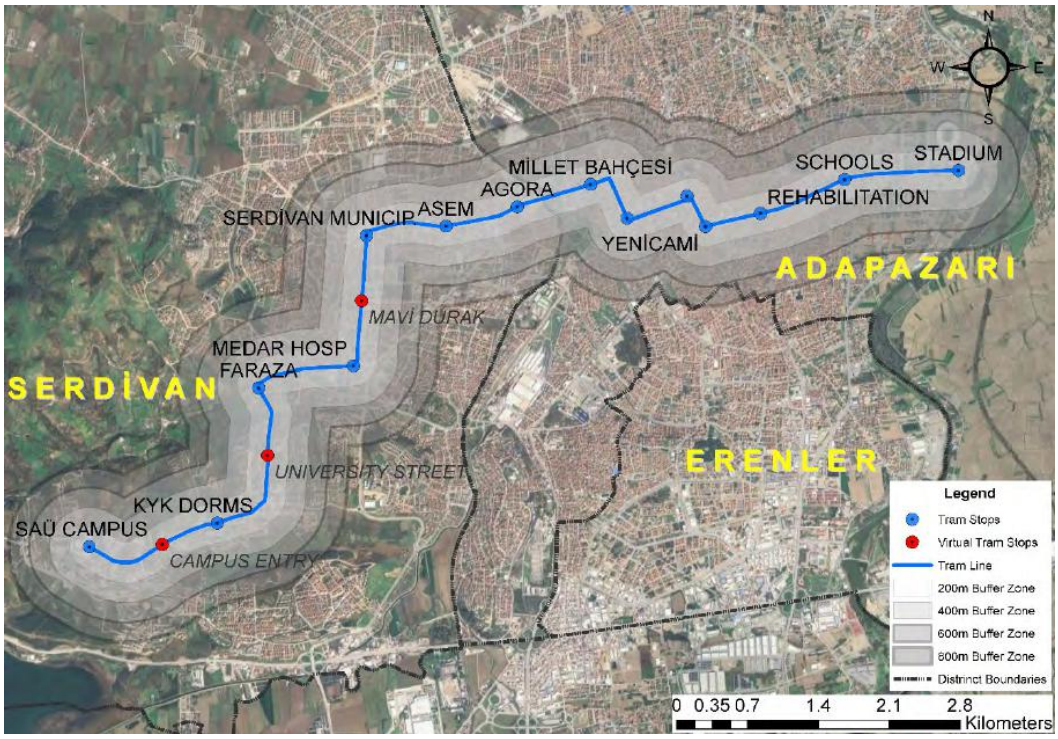

Figure 6. Buffer Area Analysis in 5,8,10 min

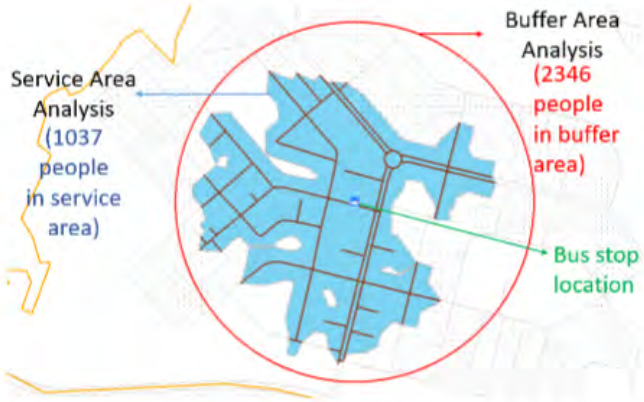

Figure 7. The difference between Service Area (Network) Analysis and Buffer Analysis in GIS (Aslan and Kocaman, 2018)

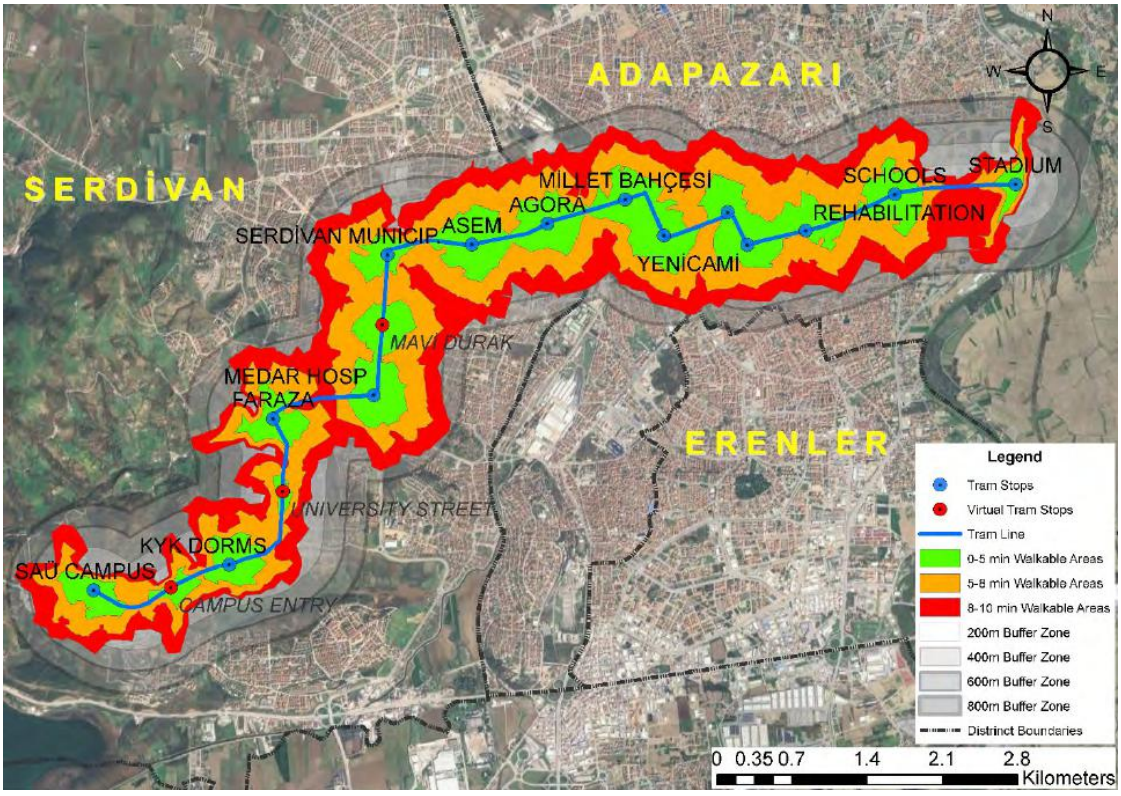

Figure 8. Time-Oriented Walking Areas and Buffer Zones on the same Map

According to Table 2, the 5-minute walkable catchment has a total surface area of 3.333 km2, the 8-minute walkable catchment has 8.213 km2, and the 10-minute walkable catchment has 12.832 km2. The buffer zones at 200, 400, 600, and 800 meters have surface areas of 4.894, 9.935, 15.125, and 20.481 kilometers, respectively. According to the 5 km/h walking speed, the walking distances for 5 minutes, 8 minutes, and 10 minutes are 416 meters, 666 meters, and 833 meters, respectively. The buffer analysis was conducted

based on the bird's eye distance, whereas the network analysis was conducted based on the walking speed. Figure 8 makes it easy to see the distinction between network analysis and buffer analysis. As previously mentioned, the buffer zone's surface areas are significantly larger than those of time-oriented zones. People cannot walk all roads and zones in urban areas because of the boundaries of personal areas or other uses, non-walkable areas like only traffic zones and some limitations for walking. Due to them, the network analysis method is more sensitive and accurate for walkability analysis.

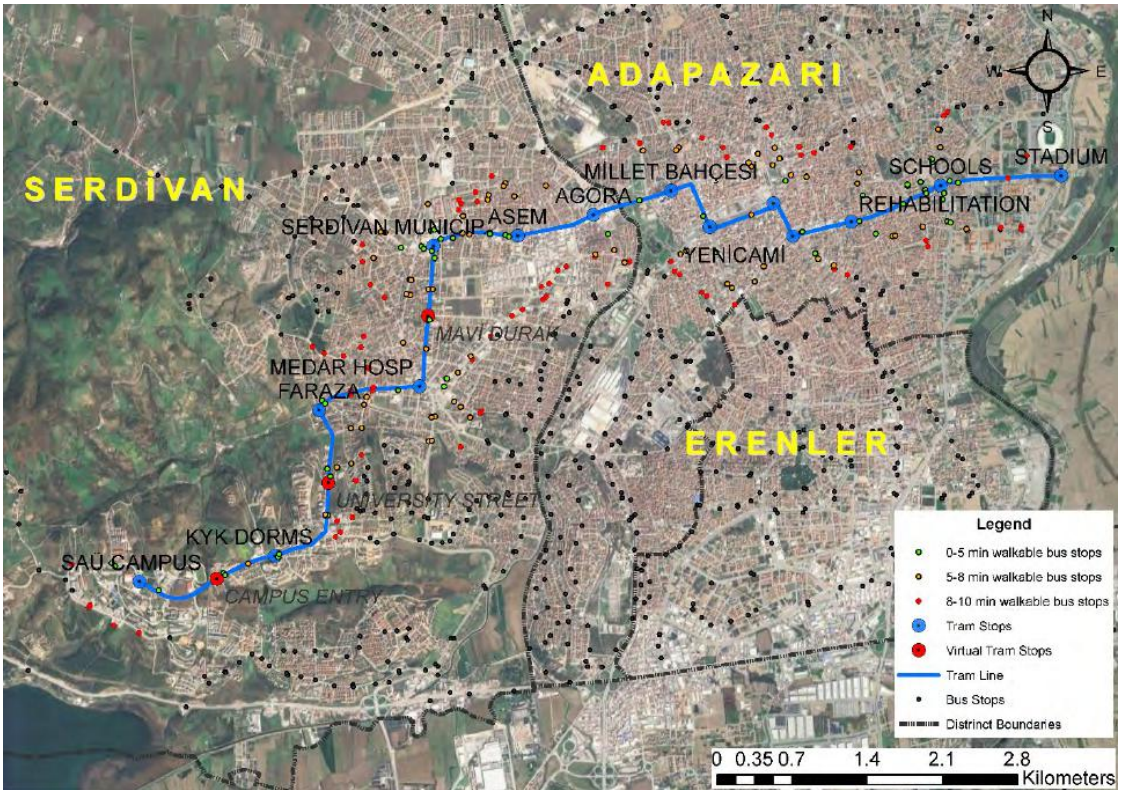

Figure 9. Bus Stops on the Map

The spatial arrangement of bus stops in respect to the suggested tram stops is depicted in Figure 9. The green markers indicate bus stations, including planned virtual tram stops, that are reachable by foot from the tram stops within five minutes. Whereas the red markers indicate bus stops within 8 to 10 minutes' walking distance from the tram stations, the orange markers suggest bus stops within 5 to 8 minutes' walking distance. After the tram line project is finished, these bus stations will be further optimized. Enhancing multimodal connection, guaranteeing smooth transitions between tram and bus services, and raising overall network efficiency are the goals of the optimization process.

Table 2. Number of stops and walking areas

	Number of Stops	Cumulative Number of Stops	Walking Areas (km^2)	Cumulative Walking Areas (km^2)
0 – 5 min	51	51	3.333	3.333
5 – 8 min	73	124	4.880	8.213

8 – 10 min	78	202	4.619	12.832

3. Discussions and Conclusions

One of the most accessible and environmentally friendly forms of transportation is walking, which provides door-to-door connection for everyday travel requirements. By offering crucial multimodal links, it enhances public transportation systems, particularly when those networks are intended to be integrated and easily accessible. A key element of urban planning is walkability, and assessing walkability can assist track and enhance the efficiency of city public transportation systems. Cities may boost public transit use, lessen dependency on private vehicles, and promote environmental sustainability by making walking a desirable alternative.

Tram stops in strategic locations, in particular, can improve public transportation's effectiveness and appeal. Tram stops should ideally be situated in places that are convenient for pedestrians to get to on foot; a distance of less than 1 km is ideal. More people use public transportation when it's close by, especially when it's safe and enjoyable to stroll. Walking distances of 416, 666, and 832 meters—or 5, 8, and 10 minutes of walking time, respectively—were examined in this study. In order to determine how far individuals are willing to walk to get tram services and how this affects the utilization of the public transportation system, these lengths were selected.

The proposed tram line in Sakarya has been developed along a visually appealing axis that crosses a range of land uses, such as workplaces, residential neighborhoods, industrial zones, educational institutions, and sports facilities. Instead of only operating during rush hour, this mixed-use development guarantees that the tram system will serve people all day long, meeting a variety of demands at various times. In addition to improving the tram's capacity use, this diversity in land use guarantees that the service will continue to be useful to a broad spectrum of consumers.

Additionally, high-interest locations like shopping malls and parks have been included in the thoughtful arrangement of the tram stops along this line. Additionally, a number of bus lines pass through or close to the tram stops, demonstrating how well-integrated these stops are with the city's current bus system. For commuters who use a variety of transportation options, this integration of modes guarantees that the stops are both handy and accessible.

A promising and convenient addition to Sakarya's transportation network, the proposed tram line combines multimodal connections, walking access paths, and thoughtfully designed tram stops. Given the high-density zones, walking areas, and effective bus line connections, it is probable that many locals will choose this new tram system as their primary means of transportation.

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